PROJECT PROPOSAL

FIA Smart ACR / PER Management System

Proposal for design, development, and on-premises deployment

Prepared For	Federal Investigation Agency (FIA)
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Prepared to support FIA's transition from manual ACR handling to a secure, trackable, role-based digital workflow for BPS 1–16 employees.

Delivery Model	Indicative Timeline	Commercial Basis	Estimated Effort
End-to-end build	**3/4 weeks**	**$15/hour**	**120 hours**

1. Executive Summary

This proposal presents a revised, client-ready approach for the FIA Smart ACR / PER Management System. The document has been updated to reflect the latest confirmed operating model: ACRs for BPS 1–16 employees are initiated by the designated clerk of the relevant office or wing, then routed to the Reporting Officer, then to the Countersigning Officer, and finally to the Secret Branch for controlled archival and retrieval.

The primary objective is to replace manual ACR movement with a secure digital workflow that improves visibility, timeliness, accountability, and record keeping. The proposed solution is designed for on-premises deployment and includes role-based dashboards, internal notifications, SMS-based alerts, overdue monitoring, priority handling, searchable archives, and complete audit trails.

Client Need	Proposed Response
Replace paper-based ACR movement with a controlled digital process.	Digitize initiation, routing, review, signing, archival, and status tracking in a single portal.

Client Need	Proposed Response
Reduce backlog and silent delays in officer queues.	Apply due dates, overdue flags, alerts, dashboards, and escalation visibility.
Create a central searchable record for all finalized ACRs.	Store final records in Secret Branch archive with indexed retrieval and audit history.

2. Confirmed Process Understanding

The revised proposal is based on the latest business understanding received from FIA and the project team. The following points are treated as the current operating baseline for proposal and architecture purposes:

- Scope covers ACR / PER handling for employees in BPS 1 to BPS 16.
- ACR initiation will be performed by the designated clerk of the concerned office or wing, not by the employee.
- The system routes each ACR from Clerk → Reporting Officer → Countersigning Officer → Secret Branch.
- Digital signatures and stamp images will be placed at predefined locations in the approved ACR form layout.
- The solution must support overdue tracking, priority ACRs, role-based dashboards, searchable records, and full logs.
- The final system will be deployed on premises within FIA-controlled infrastructure.

3. End-to-End Workflow Overview

The operational flow below captures the proposed Phase 1 process. It keeps the client workflow simple while creating a strong audit trail and better accountability for pending and overdue ACRs.

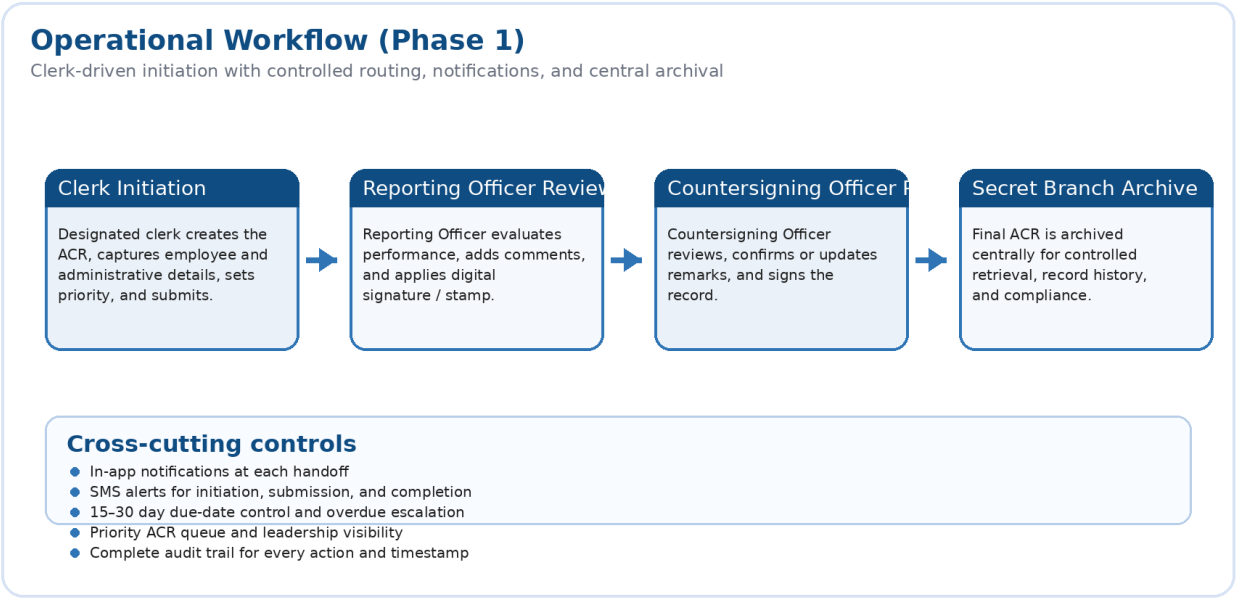

Figure 1. Phase 1 workflow from clerk initiation to Secret Branch archival.

4. Project Scope

In Scope (Phase 1)	Out of Scope / Later Phase
BPS 1–16 ACR / PER workflow automation	BPS 17+ officer evaluation automation
Clerk initiation and officer approval workflow	WhatsApp integration in Phase 1
Role-based dashboards, search, notifications, and overdue tracking	Public-facing or internet-exposed portal
Secret Branch archive, logs, and history management	Full HRMIS replacement

5. Functional Scope and Key Modules

Operational Modules	Business Outcome
Authentication and role-based access	Each user sees only the functions and records allowed for their role and wing.
Clerk initiation and data capture	Initial employee and office details are captured once and routed forward.
Reporting Officer review queue	Pending work, due dates, comments, signature, and stamp insertion in one place.
Countersigning Officer review queue	Second-stage review with approval, remarks, and final sign-off.
Secret Branch archive	Central storage, search, retrieval, and complete historical record of finalized ACRs.
Notification engine	In-app alerts and SMS notifications for handoffs, submissions, and status updates.
Priority and overdue controls	Separate tracking for urgent cases and delayed actions.

6. Role-Based Access and Dashboards

Role	Primary Access	Dashboard Focus
Clerk	Create ACR, enter base data, submit, track status	Initiated, returned, pending, and priority ACRs
Reporting Officer	Review queue, comments, sign, stamp, submit	Pending work, overdue items, average completion time
Countersigning Officer	Review, countersign, return or submit	Approval queue, overdue cases, final submission status

Role	Primary Access	Dashboard Focus
Secret Branch	Archive, retrieve, update status, controlled access	Archive count, incoming records, searchable history
Leadership / DG	Read-only oversight and analytics	Wing-wise trends, completed vs pending, overdue analysis

7. Reference Solution Architecture

The architecture below is intentionally client-friendly. It shows how the portal, workflow services, storage, identity, notifications, and optional future integration points fit together in an on-premises deployment model.

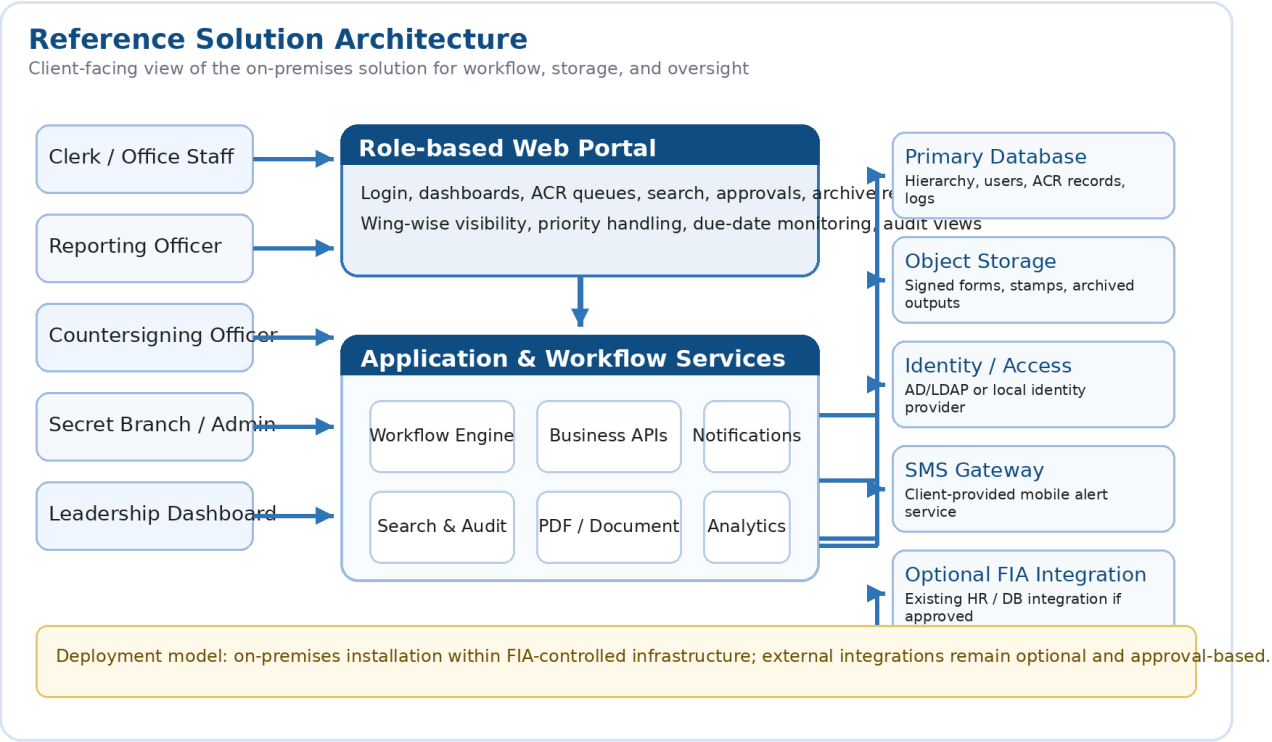

Figure 2. High-level reference architecture for the proposed on-premises solution.

Area	Recommended Technology	Purpose
Frontend	Next.js + TypeScript	Role-based portal, dashboards, forms, queues, and search interfaces
Backend	NestJS services	Business rules, approvals, archive APIs, logs, and integration layer
Workflow	Temporal or equivalent orchestration	Durable routing, timers, overdue control, and escalations
Database	PostgreSQL	Users, hierarchy, ACR records, tasks, queues, and audit data

Area	Recommended Technology	Purpose
Storage	On-prem S3-compatible storage	Signed outputs, attachments, stamp images, and archived files
Deployment	Containers / on-prem environment	Controlled internal deployment and maintainable operations

8. Delivery Plan and Timeline

For the currently understood scope, a focused three-week implementation is realistic as a controlled delivery target, provided that FIA promptly confirms form variants, hierarchy mapping, access details, and deployment prerequisites.

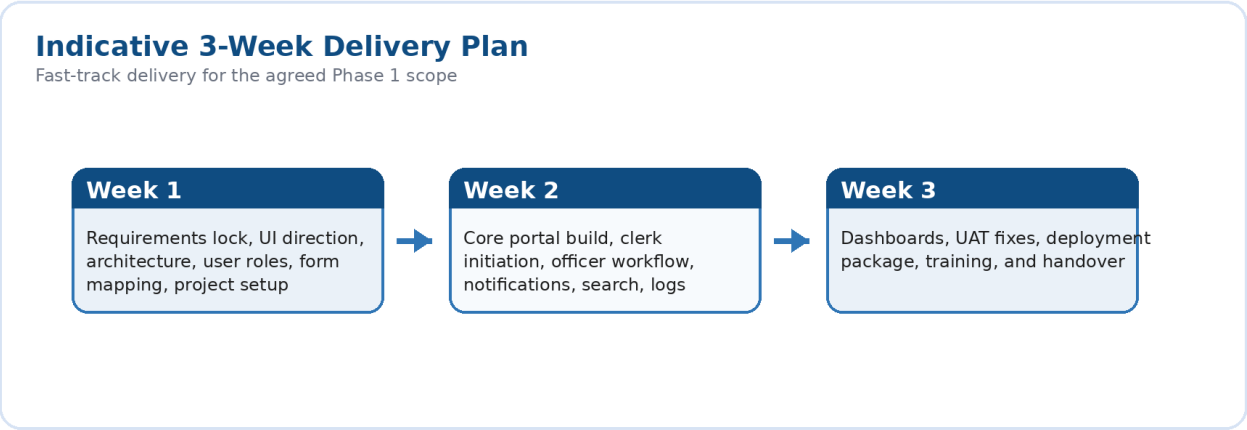

Figure 3. Indicative three-week delivery plan for the agreed Phase 1 scope.

Week	Focus	Primary Output	Client Input Needed
1	Lock requirements and design	Workflow confirmation, architecture, UI direction, form mapping	Final forms, hierarchy, policy clarifications
2	Core build	Portal foundation, clerk initiation, approval workflow, notifications, logs	Review of screenshots / demo feedback
3	Hardening and deployment	Dashboards, QA, UAT fixes, deployment package, handover	UAT participation and environment access

9. Commercial Estimate

Commercial basis: $15/hour × 40 hours/week × 3 weeks = 120 hours = $1,800±

Engagement Model	Who Provides Developers	My Role	Commercial Basis
Client-Supported Delivery	FIA provides frontend/backend developers or technical staff	Solution architecture, planning, supervision, review, coordination, delivery oversight	**Reduced consulting/management rate** — suggested as **$800 total** for oversight scope
Full Outsourced Delivery	I provide complete project team	End-to-end design, frontend, backend, database, testing, deployment, handover	**Full project rate — $1,800± total** (120 hours × $15/hour

If we go with milestone based then these are the cost breakdown.

Option	Description	Effort	Cost (USD)
Option A	Complete end-to-end Phase 1 delivery from design to on-prem deployment and handover	120 hrs	$1,800
Milestone 1	Requirements lock, solution design, architecture, and UI / workflow planning	20 hrs	$300
Milestone 2	Core portal, clerk initiation, officer workflow, dashboards, notifications, and archive features	70 hrs	$1,050
Milestone 3	Deployment support, UAT fixes, training, handover, and launch readiness	30 hrs	$450

Commercial notes:

- The above estimate is based on the requirements confirmed to date and the current Phase 1 scope.
- Any new modules, policy-driven changes, or additional integrations will be treated as a change request.
- SMS gateway costs, third-party licenses, hosting hardware, and client infrastructure are assumed to be provided or approved by FIA.
- If FIA later requests integration with existing .NET / Java systems or databases, that work should be estimated separately after technical review.

10. Assumptions, Dependencies, and Exclusions

Assumptions	Client Responsibilities	Exclusions in Current Estimate
Clerk initiates the ACR and enters the first data section.	Provide final approved ACR forms, hierarchy rules, and sample records.	No WhatsApp notifications in Phase 1.
Transfer / posting and partial-period ACR rules will follow final FIA policy.	Provide on-prem environment access, security approvals, and testing coordination.	No BPS 17+ evaluation workflow in Phase 1.
Digital stamp / signature approach used in Phase 1 is acceptable to FIA.	Participate in UAT, review cycles, and timely sign-offs.	No full HRMIS replacement or public portal.

11. Items to Lock Before Final SRS

To produce the final SRS and avoid rework, the following items should be confirmed in writing before detailed design is frozen:

Topic	Decision Needed

Topic	Decision Needed
Due-date rule	Whether the operational rule is fixed at 15 days, 30 days, or configurable by policy.
Separation events	Handling rules for retirement, resignation, death-in-service, or termination before cycle completion.
Integration scope	Whether Phase 1 remains standalone or must integrate with existing FIA data sources.
FIA Database reusability	Need time and effort to integrate/fetch the total record from other deployed system may delay the project

12. Acceptance, Validity, and Approval

This proposal is intended as a revised business and delivery proposal for client review. Once the commercial approach, workflow assumptions, and policy clarifications are accepted, the project team can proceed to finalize the Software Requirements Specification (SRS) and implementation plan.

This proposal remains valid for 30 days from the issue date unless revised in writing.

Name	Role	Signature	Date
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